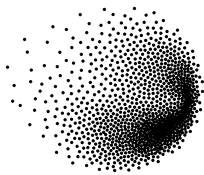




POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE



PSI

Multi-Criteria Decision Analysis for the Evaluation of Nuclear Reactor Designs

TESI DI LAUREA MAGISTRALE IN
NUCLEAR ENGINEERING - INGEGNERIA NUCLEARE

Author: **Simone Pagliuca**

Student ID: 247812

Advisor: Prof. Stefano Lorenzi

Co-advisors: Prof. Russell McKenna, Dr. Peter Burgherr, Dr. River Huang

Academic Year: 2024-25

Abstract

Here goes the Abstract in English of your thesis followed by a list of keywords. The Abstract is a concise summary of the content of the thesis (single page of text) and a guide to the most important contributions included in your thesis. The Abstract is the very last thing you write. It should be a self-contained text and should be clear to someone who hasn't (yet) read the whole manuscript. The Abstract should contain the answers to the main scientific questions that have been addressed in your thesis. It needs to summarize the adopted motivations and the adopted methodological approach as well as the findings of your work and their relevance and impact. The Abstract is the part appearing in the record of your thesis inside POLITesi, the Digital Archive of PhD and Master Theses (Laurea Magistrale) of Politecnico di Milano. The Abstract will be followed by a list of four to six keywords. Keywords are a tool to help indexers and search engines to find relevant documents. To be relevant and effective, keywords must be chosen carefully. They should represent the content of your work and be specific to your field or sub-field. Keywords may be a single word or two to four words.

Keywords: here, the keywords, of your thesis

Abstract in lingua italiana

Qui va l'Abstract in lingua italiana della tesi seguito dalla lista di parole chiave.

Parole chiave: qui, vanno, le parole chiave, della tesi

Contents

Abstract	i
Abstract in lingua italiana	iii
Contents	v
 Introduction	 1
 1 Literature Review	 3
1.1 A brief History of Nuclear Power	3
1.1.1 Origins	3
1.1.2 Generation I	3
1.1.3 Generation II	4
1.1.4 Generation III	5
1.1.5 Generation IV	5
1.1.6 Current Landscape	6
1.2 Countries	6
1.2.1 Observation on the validity of public surveys	8
1.3 MCDA	9
1.4 Relevant Studies	9
1.5 Objectives	10
 2 Preparation	 11
2.1 Criteria for Reactor Selection in Comparative Analysis	11
2.2 Country Selection for Comparative Analysis	15
2.3 Attribute Definition	15
2.3.1 Technical parameters	15
2.3.2 MOX Fuel Compatibility	16
2.3.3 Technological Readiness	17

2.3.4	Safety	17
2.3.5	Capital Costs	17
2.3.6	Commissioning Time	18
2.3.7	Operational Costs	19
2.3.8	Potential Impact on the energy market	19
2.3.9	Potential Impact on greenhouse gas emissions	20
2.3.10	Waste	20
2.3.11	Size - Nominal output power	20
2.3.12	Load Following Capabilities	20
3	Analysis	23
4	Results	25
	Bibliography	27
A	Appendix A	39
B	Appendix B	41
	List of Figures	43
	List of Tables	45
	List of Symbols	47
	Acknowledgements	49

Introduction

The global energy landscape is undergoing a significant transformation, driven by the imperative to decarbonize electricity generation while ensuring energy security and system reliability. Within this context, the European energy transition is accelerating and nuclear power can play a critical role to both provide low carbon electricity and stabilize the grid amid the increase use of renewable power sources.

National requirements in terms of energy vary significantly, while some prioritize baseload capacity to offset import dependence, others require flexible solutions to be coupled with existing and future renewables. The emergence of Small Modular Reactors alongside established Generation III+ designs expands possibility but also introduces complexities to the decision making process. The commitment to a specific nuclear reactor design involves complex tradeoffs across technical, economic, regulatory, social, environmental, and geopolitical dimensions, which economic, social and political consequences will last for decades.

This thesis evaluates selected nuclear reactor designs for potential European deployment by employing Multi-Attribute Value Theory (MAVT), a structured decision-support methodology, and examines how different national contexts may influence the assessment process.

The research addresses four key questions:

- Which reactor designs are viable for European deployment based on geopolitical issues, technology maturity, and market availability?
- What are the key technical, economic and social attributes that differentiate the viable reactor designs?
- How do the energy requirements and socio-political contexts of different European countries influence these attributes?
- How does the application of a Multi-Attribute Value Theory (MAVT) framework reveal differences in reactor design rankings across countries?

1 | Literature Review

1.1. A brief History of Nuclear Power

1.1.1. Origins

The 1930s and 1940s witnessed rapid advancement in nuclear science. In 1932 James Chadwick discovered the neutron [1] and a year later Leo Szilard hypothesized fission chain reactions. In 1938, Otto Hahn and Fritz Strassmann produced barium by bombarding uranium with neutrons, demonstrating nuclear fission [2]. Further advancements like the first controlled nuclear chain reaction in the Chicago Pile-1 achieved by Enrico Fermi's team in 1942, were speeded up by World War II as part of the Manhattan Project which led to the atomic bombings of Hiroshima (6 August 1945) and Nagasaki (9 August 1945). The recognition among the scientific community of the peaceful potential of nuclear technology led to the famous speech "Atoms for Peace" by USA President Eisenhower in 1953 to the United Nations [3] which established the foundation for international nuclear cooperation. The International Atomic Energy Agency (IAEA) and the European Atomic Energy Community (EURATOM) were established 4 years after, in 1957, and were tasked with promoting peaceful nuclear applications while serving as independent authorities for safety regulation and non-proliferation oversight.

1.1.2. Generation I

The 1950s marked the emergence of numerous reactor designs, driven by both civilian energy needs and military applications. These early models, now classified as Generation I reactors, represent the first prototypes of nuclear power at utility scale. The Experimental Breeder Reactor I (EBR-I) in Idaho (USA), which first generated electricity in 1951 (powering four light bulbs). The Obninsk Nuclear Power Plant (USSR), which in 1954 became the first to feed approximately 5 MWe into a local power grid. Calder Hall-1 (UK), the world's first commercial-scale nuclear power station, which began operation in 1956 with a 50 MWe capacity connected to the national grid.

The United States pioneered light water reactor (LWR) technology through two distinct designs: the Boiling Water Reactor (BWR) and the Pressurized Water Reactor (PWR) which was originally developed for naval propulsion. The first commercial implementations of the BWR was the Dresden-I (1960) and for the PWR the Shippingport Atomic Power Station (1957) [4]. Concurrently, the USSR developed its own pressurized water technology, the VVER ("Water-Water Energetic Reactor"), featuring design characteristics that persist in modern variants.

This period also witnessed significant technological diversification as other nations developed their own reactor designs. The United Kingdom advanced gas-cooled Magnox reactors, optimized for dual electricity and plutonium production. The Soviet Union initiated development of the graphite-moderated, water-cooled RBMK design, while Canada's nuclear program focused on pressurized heavy water reactor (PHWR) technology, culminating in the first commercial CANDU reactor at Douglas Point in 1968 [4].

1.1.3. Generation II

Between the 60s and 70s Generation II reactors were deployed, taking advantage of the experience from the previous run, these reactors are not anymore considered experiments or prototypes but fully commercial power plants. In these decades numerous countries commissioned their first nuclear reactors: Belgium (1962), Italy (1963), Japan (1963), Sweden (1964), Switzerland (1969), Slovakia (1972), Argentina (1974), Bulgaria (1974) and Armenia (1976) [4]. Following the 1973 oil crisis, the deployment of nuclear power plants was accelerated in some countries as an energy security strategy. Most notably France implemented the Messmer Plan, named after the Prime Minister at the time, to greatly expand the nuclear plant fleet [5].

Nuclear expansion continued through the 1980s until 1986. As news from the Chernobyl disaster came to western countries, fear of the potential risks of nuclear power intensified. The public opposition grew significantly so that, even if safety and regulatory updates were implemented, several countries decided to slow down, reduce or halt their nuclear programs altogether, such was the case of Italy after the results of the 1987 referendum, which resulted in a complete phase-out of nuclear power [6, 4].

By the 1990s, multiple western countries had suspended nuclear deployment entirely, including Germany and Switzerland. France, previously the most active builder, connected 10 reactors between 1990-1999, while the United States and United Kingdom only deployed 4 and 1 reactors respectively during the same period. Italy completed its nuclear phase-out by 1990 with the permanent shutdown of all operating reactors [4].

Contrastingly, Eastern nations pursued an opposite trajectory. Following the post-Soviet economic transition, Russia resumed its nuclear program and has since continued expansion of its reactor fleet [4]. China initiated its nuclear program in the 1990s with Qinshan-1 (connected in 1991) and has since maintained aggressive expansion. Over the past three decades, China has brought 56 additional reactors online and currently has 29 under construction, positioning itself to surpass both France and the United States in total reactor capacity within the coming decade [4].

1.1.4. Generation III

As a result of the regulatory safety improvements, a stabilized choice of technology with LWRs settling as the to go choice all over the world and even larger capacities, Generation III reactor started to be developed in the 90s.

The improvement of these designs was fueled in the early 2000s by concerns over climate change, energy security, and rising fossil fuel prices, leading to Generation III+ reactors, such as the European Pressurized Reactor (EPR) or the US AP-1000 which improve safety and reliability in a consolidated plant layout. However after the 2011 Fukushima Daiichi accident Western nations experienced a significant setback: Germany accelerated its nuclear phase-out policy, while Italy reaffirmed its anti-nuclear position through a popular referendum [7]. In the following years projects faced substantial delays, cost overruns, and public opposition, with France's Flamanville-3 EPR serving as a particularly illustrative case experiencing budget increases to more than four times original estimates and construction timelines tripled from initial projections [8].

1.1.5. Generation IV

From the late 2000s to nowadays, new nuclear commissioning projects can be classified as "Mega Projects", such term is used to refer to very large commitments where economies of scale tend to fail due to the complexity of the task at hand, resulting in massive delays and cost overruns, well known examples of Mega Projects are Suez and Panama Canals, the Sydney Opera House or the development of the Concorde Supersonic Aeroplane [9, 10].

To counter this phenomena, the nuclear industry is developing Small Modular Reactors (SMRs) which are part of Generation IV reactors. SMRs are smaller, standardized reactors that can be built mostly off site to take advantage of economies of multiples. Some SMR designs are based on novelty technologies while others take advantage of well known water-based designs.

Other generation IV reactors are being design to tackle on other aspects of nuclear power such as waste minimization or higher efficiencies. Manly by employing different means of cooling to exploit neutrons at higher energies (fast neutrons), or by using thorium instead of uranium as fuel or even using direct gas based cycles to increase thermal efficiency.

1.1.6. Current Landscape

This historical overview demonstrates the complexity behind the decision to adopt nuclear power which since it's beginning has been a matter of strategic geopolitical and historical consideration rather than purely technical or economic factors. So for starters we observed the current situation and consideration of nuclear energy in europe:

In 2020, the COVID-19 pandemic exposed vulnerabilities in global economic systems and Russia's invasion of Ukraine in 2022 further revealed Europe's energy dependencies [11, 12, 13]. At the same time, decarbonization goals to combat climate change have prompted Western nations to reconsider nuclear energy as a valid power source able to provide low-carbon, reliable power while enhancing energy independence and system resilience [14].

This strategic reassessment of nuclear is evident in recent policy initiatives. The European Union's classification of nuclear power as a sustainable investment under its taxonomy for climate mitigation [15], alongside programs like the U.S. Department of Energy's Advanced Reactor Demonstration Program [16], are a clear indicator of the renewed interest in nuclear technologies.

1.2. Countries

It is important to note that nuclear energy projects are charaterized by long development timelines from initial planning to commercial operation and can therefore face unique challenges related to political and social continuity. Frequent policy changes may introduce delays, require costly redesigns or halt projects althogether. For this reason the present study has considered four european country with different energy and political landscapes. Moreover, even if the present study considers only european countries, their demand profiles in terms of electricity are all but similar. We are now going to introduce the four countries that were chosen for this study, presenting their current poiltical and popular opinions on nuclear observed by surveys, news and goverment communications as well as the charateristics of their electricity market, which data has been manipulated from the dataset of the ENTSOE Transparency Platform [17].

France demonstrates consistent public support for nuclear energy, with 75% favoring continued operation of existing plants and 65% supporting new construction according to a 2022 IFOP survey [18]. Given the long-standing consensus on nuclear power as a strategic energy pillar and the country's established industrial base, significant policy reversals appear unlikely in the near term. The country electric system relies heavily on nuclear power, which accounted for 67% of its electricity generation in 2024 [19]. Despite the substantial demand, which averaged 49.05 ± 9.83 GW in 2024, France is a net electricity exporter, with 21% of total domestic production of electricity being exported in 2024.

Switzerland presents a more complex picture. Following the 2011 Fukushima accident, the government adopted a phase-out policy later confirmed by a 2017 referendum (42% approval) [20]. However, recent surveys indicate shifting public opinion, with 53% support [21, 22, 23]. The federal government has subsequently signaled potential reconsideration of nuclear's role in meeting climate targets [24], though the narrow margin of support and Switzerland's direct democracy system create policy uncertainty.

The Swiss grid is characterized by relatively low demand (6.79 ± 1.12 GW on average in 2024) and generation dominated by renewables (65% of total generation), particularly hydroelectric which accounted for 59% of the electricity generated in 2024 [19]. Such a renewable intensive generation mix results in uneven cross border flows which average 2.71 ± 3.14 GW. The large standard deviation is a consequence of periodic overproduction requiring exports, alternating with imports whenever renewable generation is low.

Poland shows strong public backing for nuclear development, with recent surveys indicating up to 92% support [25]. This public sentiment aligns with progress in the country's nuclear program, such as preliminary site works and vendor agreements [26, 27], suggesting firm governmental commitment to nuclear deployment. The Polish electricity sector is heavily coal dependent, being responsible for producing 55% of its electricity in 2024 [19]. Such reliance on coal positions Poland as the country with the highest green house gas emission intensity in Europe. [28].

Italy's nuclear landscape remains constrained by the 1987 post-Chernobyl referendum and its 2011 reaffirmation following Fukushima, which mandated complete phase-out and new construction ban respectively. However, since the referendum in Italy are abrogative and do not need to another popular vote to change direction, a government decree is enough to reintroduce nuclear, which is what the current government has plans for [29]. Italy's history of political volatility and the long-term nature of nuclear projects create significant implementation challenges regardless of public opinion trends.

This study focuses exclusively on Northern Italy due to the country's fragmented electricity market structure. The Northern bidding zone represents the energy intensive of the country, with an average load of 17.66 ± 4.35 GW and significant import dependence. Cross-border flows occur 99% of the time in a year to transfer on average into the region 5.23 ± 1.67 GW.

1.2.1. Observation on the validity of public surveys

While public opinion surveys could provide, in principle, quantitative measures on qualitative matters such as support for nuclear installations, closer examination often reveals methodological limitations that may skew results. In Italy, a 2024 IPSOS survey commissioned by Legambiente (an environmental NGO with a known anti-nuclear position) reported 81% opposition to nuclear power [30]. However, analysis of the survey report reveals problematic question design. The primary question was phrased as: *"The debate on nuclear energy has returned to the forefront. What is your opinion on the matter?"* with response options structured as:

- A1: "The current situation is complex; a return to nuclear should be evaluated"
- A2: "Nuclear has more risks than benefits and is not the right path"
- A3: "Nuclear energy entails excessive hidden costs and is not advantageous"
- A4: "Only safer technologies than current ones should be considered"

This framing demonstrates clear bias: two options (A2 and A3) explicitly oppose nuclear, A1 offers only conditional support, and A4 introduces safety concerns before suggesting potential support. Notably, the 81% opposition figure includes A4 responses, which represent conditional rather than absolute opposition. When considering only explicit opposition (A2+A3), the figure decreases to 43%.

A 2024 SWG study presents a more methodologically sound approach, though its conference presentation emphasized positive findings. The actual data reveals a more nuanced picture: over 80% of respondents consider themselves insufficiently informed about nuclear topics, and further investigation found that approximately 90% were unaware of SMRs or Generation IV reactors [31]. These findings suggest that public opinion on nuclear energy may be more influenced by lack of information than by firmly held positions.

1.3. MCDA

The decision making process begins by identifying the decision problem and the possible solutions to such problem. Then, analytical frameworks, usually referred to as decisional support tools, have been developed to assist decision makers in navigating such decision problems. A common approach to decision making, which is encountered by anyone daily in day to day life, is to look at the possible solutions or "alternatives", compare positive and negative outcomes and find the best, or less worst, between them.

A more advanced method is Multi-Criteria Decision Analysis (MCDA) which first breaks down alternatives into attributes (consequences of the alternative or traits that distinguish alternatives), and then analyzes tradeoffs between attributes rather than between alternatives [32]. While there exists many MCDA methodologies (see Greco et Al. 2016 [33] and Cinelli et al. 2020 [34]) the common ground is to have a model of the decision problem that contains the information about the problem itself, considers trade-offs and allows to choose between alternatives [35].

As just presented, in MCDA the decision problem has to be broken down into smaller objectives or criteria. Each criterion is evaluated by means of an indicator, which can be of quantitative or qualitative origin, might be derived from measurements, estimates, probabilistic assessment, expert opinions, public surveys or a combination of these.

1.4. Relevant Studies

A review of relevant literature reveals numerous applications of MCDA methods across diverse fields. Focusing specifically on studies combining MCDA with nuclear power decision-making, several distinct approaches emerge:

Site selection studies represent the most common application, typically employing GIS-based software to evaluate potential locations for new nuclear installations within limited geographical regions. These analyses frequently incorporate quantitative non-technical factors such as population proximity [36] [37]. Abdullah et al. (2025) extend this approach by including social dimensions like public acceptance and tourism impacts, which present challenges for quantification [38]. Notably, Devanand et al. (2019) develop an algebraic optimization model to identify optimal SMR locations in Singapore, though like most site-focused studies, they favor a design without systematic comparison [39].

A second category compares different nuclear technologies. Kiser and Otero (2023) evaluates PWRs, SMRs, and Molten Salt Reactors [40], while Locatelli and Mancini (2011)

compares large and small reactor designs [41]. Birikorang's study stands out for its regional focus on South Africa and comparison of three specific reactor models (NuScale SMR, EM2-HTGR, FTR-MSR), though its technological comparisons mix designs at different maturity levels [42]. Notably, all three studies utilize the Analytic Hierarchy Process (AHP) or its Fuzzy variant. The last one (Birikorang) draws on the IAEA's INPRO methodology for advanced nuclear technology comparison [43].

Broader energy comparison studies by Hernández-Torres (2025), Pasaoglu (2018), and Atilgan (2016) evaluate multiple generation options (typically eight alternatives) employing AHP-TOPSIS, AHP, and MAVT methodologies respectively. These studies use generic, averaged parameters for each technology class [44, 45, 46]. In contrast, Castellazo (2014) adopts a more precise approach using Multi-Attribute Value Theory (MAVT) to compare thirteen energy technologies, employing specific reference plants (e.g. EPR reactor for nuclear power) rather than generalized technology averages [47].

Three studies show particular relevance to the current research. Zolkaffly et al. (2014) evaluates nuclear plant models in Malaysia but suffers from outdated data and neglects non-technical aspects [48]. Topaloğlu (2025) applies the Analytic Network Process (ANP) to compare reactor types (PWR, BWR, PHWR) and locations, though social considerations remain limited to site-specific factors [49]. Goushis (2025) employs TOPSIS to assess SMR deployment in the EU for coal replacement, but focuses on a single reactor model rather than comparative technology evaluation [50].

The literature review reveals three gaps in nuclear energy decision-making research. First, most studies focus solely on site selection rather than comparing different nuclear technologies. Second, comparisons between nuclear designs lack consideration of a specific decision-making context. Third, studies that do consider specific decision-making context, examine various energy generation options as broad technology categories (e.g., nuclear, solar, wind) rather than conducting detailed analyses of specific reactor models or power plant designs.

1.5. Objectives

DRAFT - Da controllare una volta finito di scrivere This study will consider real world data, literature estimates, industry claims and expert judgment to rank six nuclear power plant designs in four different european national contexts. In the first section the selection process of reactors to be evaluated is presented. In the second section the definition of indicators, value functions and weights is discussed. The third section is dedicated to the final discussion

2 | Preparation

2.1. Criteria for Reactor Selection in Comparative Analysis

This study focuses on a curated set of reactor designs at an advanced stage of development. The selection process begins with the the IAEA "Nuclear Reactors in the World" [51], which catalogs operational, under-construction, and decommissioned reactors globally.

The initial selection process applied two primary exclusion criteria. First, designs were eliminated based on supplier origin, removing all reactors associated with countries unlikely to engage in nuclear technology transfer with Europe under the current geopolitical climate. This filter specifically excluded reactors developed in or supplied by Russia, China, and other nations maintaining restricted technology-sharing agreements with Western markets.

Second, reactors commissioned before 2000 were excluded from consideration, as their designs represent outdated technological standards. The complete filtering methodology is documented in the accompanying analysis notebook (PRIS Data Analysis.ipynb), which details how the initial dataset was reduced to nine distinct reactor models.

From this pre-selected group, three designs - the EPR-1750, APR-1400, and AP-1000 - were prioritized for comprehensive analysis. It should be noted that while the EPR (1650 MWe) and EPR-1750 share significant design similarities, they are treated as separate models in this study. Additionally, the Advanced Boiling Water Reactor (ABWR), though listed as operational in the IAEA ARIS database, was excluded following verification through the PRIS database, which indicates all ABWR units have suspended operation since 2011 [4]. The selection process and its outcomes are summarized in Table 2.1.

Subsequently, the IAEA Advanced Reactors Information System (ARIS) database [52], was consulted to identify advanced reactor models classified as under design, under reg-

ulatory review, under construction, or operational. Demonstrations units and prototypes were excluded from this set, yielding 10 candidate designs. To avoid redundancy, models already included in the prior selection (e.g., Generation III+ designs captured in both datasets) were removed. Each remaining design was then assessed for development status, technological readiness, and applicability to European energy systems as summarized in Table 2.2.

The overall selection produced a set of six reactors, three Generation III+ and three SMRs. Their main characteristics are summarized in Table 2.3. The most notable differences are the integrated designs of the BWRX-300 and Nuscale SMR which eliminate the external primary loop in favour of steam generator inside the reactor pressure vessel. This allows for compact construction and allows to exploit natural circulation for cooling. Then the Nuscale SMR is the only model designed to be deployed in a multi unit configuration, with up to 12 modules in the same containment building, for this study the 6 module plant (VOYGR-6) has been selected as it recently recived the Standard Design Approval by the U.S. Nuclear Regulatory Commission in May 2025. "A standard design approval indicates that a proposed reactor design meets applicable agency safety requirements. Companies that seek to use the US460 design would have to file applications seeking permission to build and operate a nuclear reactor using the approved design" [59].

Maybe here add a paragraph for each design to explain their main features.

Reactor Name	Country	Technology-sharing	Commercial Operation	Up Date	to Decision
APR1400	S.Korea	✓	✓	✓	Selected
CAREM Prototyp	Argentina	✓	✗	✓	Rejected
PRE KONVOI	Germany	✓	✓	✗	Rejected
EPR 1750	France	✓	✓	✓	Selected
AP1000	USA	✓	✓	✓	Selected
ABWR	Japan	✓	✗	✓	Rejected
APWR	Japan	✓	✗	✓	Rejected
EPR	France	✓	✓	✗	Rejected
OPR-1000	S.Korea	✓	✓	✗	Rejected

Table 2.1: Selection of reactor models for analysis

Reactor Name	Country	Used	Reasoning
APR+	S.Korea	✗	As reported in the manufacturer website [53] this design is mean to be an "indigenous reactor cleared of obstacles to export"
ATMEA	Japan, France	✗	Joint venture was abandoned in 2019 [54]
BWRX300	USA, Canada	✓	Design from GE-Hitachi, a lot of support from US and Canada as well as interes in the EU [55] [56].
i-SMR	S.Korea	✗	Never expressed interest in development outside south Asia and the middle east, no updates since early 2024.
iMSR400	US	✗	No interest overseas, project seems to be at an early stage of development.
Natrium	US	✗	Demo plant [57].
Nuward	France	✗	After the change in direction of 2024 the project has been set back [54].
Nuscale	US	✓	Even if there was some financial controversy in 2024 [54]. the project seems to be still moving forward in terms of goverment founding [54] and regulatory review [54].
PRISM	US	✗	GE-Hitachi considers it a "concept ... currently being put into practice in two reactors: Natrium and ARC-100" [54].
Rolls Royce SMR	UK	✓	It's moving forward and from the technological point of view this is the simples as it it litteraly a scaled down PWR[54].
SMART	S.Korea, Saudi Arabia	✗	Plan to export only to south-east asia and middle east [58].

Table 2.2: Selection of advanced reactor models for analysis

Reactor Name	Type	Construction	P_e (MWe)	Supplier
APR1400	PWR	1 Unit	1340	KEPCO (S.Korea)
EPR 1750	PWR	1 Unit	1630	Framatome (France)
AP1000	PWR	1 Unit	1120	Westinghouse (USA)
BWRX300	i-BWR	1 Unit	300	GE-Hitachi (USA-Canada)
Rolls Royce SMR	PWR	4 to 12 Units	470	Rolls Royce (UK)
Nuscale	i-PWR	1 Unit	77	Nuscale Power (USA)

Table 2.3: Final selection of reactor models for analysis

2.2. Country Selection for Comparative Analysis

This study evaluates four European countries: France, Italy, Poland and Switzerland, to assess how national energy contexts influence the ranking of nuclear reactor designs.

In Italy, the energy market is divided into different zones, each with its own demand patterns. For this analysis, Northern Italy serves as the focal region being the country's most industrialized and energy-intensive area[54]. It is characterized by exceptionally high electricity demand, a strong reliance on imports, and above-average greenhouse gas emission intensity, making it a critical bottleneck in Italy's decarbonization efforts. Additionally, the country's historical political volatility introduces uncertainty into long-term energy planning.

France, with $\sim 70\%$ of its electricity generated by nuclear power[54], represents a mature nuclear market. Here, the marginal gains from additional reactors are likely minimal, offering insight into how reactor rankings shift in a saturated nuclear energy landscape.

Poland, heavily dependent on coal and among Europe's highest GHG emitters [54], presents an opposite case. As the country seeks to modernize its grid and reduce emissions through nuclear energy [54], it provides an opportunity to assess which reactor designs could best support its transition while meeting growing energy demand.

Finally, Switzerland provides a unique case due to its low electricity demand, hydropower-dominated grid, and prudent nuclear policy. Its constrained grid capacity and emphasis on energy independence make it ideal for assessing small modular reactors (SMRs) and their role in complementing renewables.

2.3. Attribute Definition

Throughout this section we will define the indicators.

2.3.1. Technical parameters

...

2.3.2. MOX Fuel Compatibility

Mixed Oxide (MOX) fuel is produced through spent nuclear fuel reprocessing and offers a solution for partially closing the nuclear fuel cycle in thermal reactors. While this process generates some secondary waste, it provides several strategic advantages: reducing final nuclear waste volumes, minimizing mining waste, and decreasing dependence on uranium imports. The latter is particularly relevant for Europe, which lacks significant domestic uranium deposits. However, reprocessing also presents proliferation risks since the extracted plutonium could potentially be diverted for weapons production. Due to these competing factors, nuclear fuel reprocessing is a fundamental strategic decision; the United States has foregone this practice, while France and UK have implemented the process in their nuclear programs [60].

Given the European context of this study, MOX compatibility is considered a relevant reactor attribute. The evaluation methodology assigned each reactor design an initial score (0-10) based on its maximum MOX core fraction capability, followed by qualitative adjustments reflecting design-specific features. The complete scoring process and results are detailed in Table 2.4.

Reactor Model	MOX[%]	Considerations	Score	Sources
APR1400	33	-	3/10	[52]
EPR 1750	100	50% without modifications, to reach 100% "design adaptations which are not significantly costly" are required.	9/10	[52, 61]
AP1000	50	50% without modification, to reach 100% major modifications are required	6/10	[52, 62]
BWRX-300	0	GNF-2 Not manufactured with MOX	0/10	[63]
Rolls Royce SMR	0	Rolls Royce states that they are not planning to use MOX	0/10	[64]
NuScale	0	Not licensing for MOX but a study reveals support is possible without design modifications	1/10	[65, 66]

Table 2.4: MOX fuel compatibility scores and maximum MOX fraction

2.3.3. Technological Readiness

Expert consultation

2.3.4. Safety

To be discussed

2.3.5. Capital Costs

DRAFT

Option	PROs	CONs
Use NOAK estimates	Use of estimates in all cases, comparable values	It's not realistic and not fair towards more mature designs
Use FOAK estimates/data	Better data for three large reactors	We are mixing up real data and estimates
Use NOAK and apply FOAK-to-NOAK conversion	Comparable values, with awareness of current know-how	rely on a model
Present NOAK and FOAK to experts along with technological readiness overview and let them evaluate	As before	rely on expert judgment

Table 2.5: Options overview

The same applies to commissioning time.

It is important to note that these costs refer to an Nth of a kind installation and will not be representative of the FOAK costs. Moreover as detailed by Rothwell 2022 [67] the costs of recent nuclear installations in Europe and the US have been significantly higher than those announced at the start of the project. For this reason we will apply a FOAK-to-NOAK factor to better represent the real costs. For example the EPR-1600 announced costs for Flamanville-3 was $1886 \text{ USD}_{2020}/kW_e$ but have risen to $8600 \text{ USD}_{2020}/kW_e$ [68]. All Values are summarized in Table 2.6.

For reference, the capital costs of the same plants as the previous section have been

reported by the US Energy Information Administration [69].

We want to clarify that the data has all been converted to 2020 USD since the main source of data [70] and [68] reports all costs in this currency and year.

Reactor Model	Capital Cost of NOAK [\$/kWe]	Capital Cost of FOAK [\$/kWe]	Data Source
APR-1400	1828	2410	[70, 68]
EPR-1750	2150	8600	[70, 68]
AP-1000	2000	8600	[70, 68]
BWRX-300	2318	3467	[71]
Rolls Royce SMR	6050	7408	[72]
NuScale	2850	3600	[73]
Other Power Plants			
Onshore Wind	-	1265	[69]
Solar PV + Battery	-	2175	[69]
Hydroelectric	-	6007	[69]

Table 2.6: Capital costs for selected reactor models

2.3.6. Commissioning Time

DRAFT

See observations in the previous section

I would apply a +0.5 years to planned commissioning time on the APR and AP based on the planned commissioning time of the EPR that differs by 0.5 between UK and Chinese plannings.

Reactor Model	Planned Commissioning Time	Commis- sioning Time	Last unit Commissioning Time	Data Source
APR1400	5.0 in Korea		10 and 8 in Korea	[8]
EPR 1750	5.0 in Europe, 4.5 in China		9 in Taishan, 17 in Flamanville and Olkiluoto	[8]
AP1000	5.0 in USA		9 in Sanmen, 10 and 11 at Vogtle	[8]
BWRX-300	2 ~ 3		NA	[74]
Rolls Royce SMR	4.0		NA	[75]
NuScale	3.0		NA	[76]

Table 2.7: Technological readiness and planned commissioning time

2.3.7. Operational Costs

Lascks presentation of the attribute

Steigerwald et al. (2023) [70] provides comprehensive cost analyses for future nuclear installations, including capital and operational expenditures for both SMRs and large reactors. The study reports operational costs in USD_{2020}/MWh_e for EPR and APR1400 designs, as well as a "Long-Term US" category which this analysis uses as AP1000 costs. For SMRs, operational costs were originally presented in power-based units (USD_{2020}/MW_e), requiring conversion to energy-based measurements (USD_{2020}/MWh_e) using a Gaussian distribution of capacity factors ($\mu = 90\%$, $\sigma = 5\%$) according to:

$$\frac{USD_{2020}}{MWh_e} = \frac{USD_{2020}}{MW_e} \cdot \frac{1}{CF \cdot 8760 \frac{h}{year}}$$

The same study reports fuel costs of $6.23USD_{2020}/MWh_e$ for PWR-based technologies with the exceptions of NuScale at $7.15USD_{2020}/MWh_e$ and the only BWR reactor at $29.55USD_{2020}/MWh_e$. For comparative context, operational cost estimates for utility-scale onshore wind, hydroelectric, and solar PV (with single axis tracking and DC battery storage) were obtained from the U.S. Energy Information Administration [69]. All cost data have been standardized to 2020 USD to maintain consistency with the primary data sources [70, 68], values are presented in Table 2.8.

Reactor Model	Operative Cost [\$/MWh]	Data Source
APR-1400	18.44	[70]
EPR-1750	14.26	[70]
AP-1000	18.69	[70]
BWRX-300	18.30 ± 0.82	[70]
Rolls Royce SMR	65.68 ± 2.94	[70]
NuScale	22.76 ± 1.02	[70]
Other Power Plants		
Onshore Wind	28.08	[69]
Solar PV + Battery	33.33	[69]
Hydroelectric	28.49	[69]

Table 2.8: Economic costs for selected reactor models

2.3.8. Potential Impact on the energy market

To write

2.3.9. Potential Impact on greenhouse gas emissions

To write

2.3.10. Waste

To write

2.3.11. Size - Nominal output power

To write

2.3.12. Load Following Capabilities

Load-following capability can be quantitatively assessed using characteristic curves that represent ramp-up and ramp-down rates as a function of load (expressed as percentage of nominal power $\%P_{nom}$), as illustrated in Figure 2.1. To facilitate comparative analysis, these performance curves are reduced to scalar values through integration of the area under said curve.

For benchmarking purposes, load-following capabilities were calculated for two modern gas turbine configurations: A heavy-duty combined cycle plant featuring two GE Vernova 9HA.02 gas turbines (2015 data) [77] and an equivalent plant configuration utilizing aeroderivative GE LM2500XPRESS+G5 DLE UPT gas turbines (2025 data) [78]. Hydroelectric plants are also included as a reference for the highest flexibility, a 1989 publication by the US Office of Technology Assessment [79] reports that the response rate of large ($>60\text{MW}$) hydroelectric plants ranges from 4 to 6 $\%P_{nom}/\text{s}$, which translates to 240 to 360 $\%P_{nom}/\text{min}$. A graph from a 1994 conference reported in a technical analysis from 2009 shows that some hydro plant can achieve 0 to 100% in much less than a minute and from black startup to full power in around 5 minutes For the comparison we indicate greater than 100%. [80, 81]

To establish an upper benchmark hydroelectric plants were included in the comparative analysis. A 1989 U.S. Office of Technology Assessment report [79] documents that large ($>60\text{ MW}$) hydroelectric facilities demonstrate remarkable load-following capabilities, with response rates ranging from 4 to 6 $\%P_{nom}/\text{s}$ (equivalent to 240-360 $\%P_{nom}/\text{min}$). Additional performance data from a 1994 conference presentation, cited in a 2009 technical analysis [80, 81], indicates that certain hydroelectric installations can achieve 0-100% load changes in under one minute and black start to full power operation in approximately five

minutes. For the purposes of this comparison, hydroelectric flexibility is conservatively represented as $>100\%P_{nom}/\text{min}$.

The results of these analyses are presented in Table 2.9.

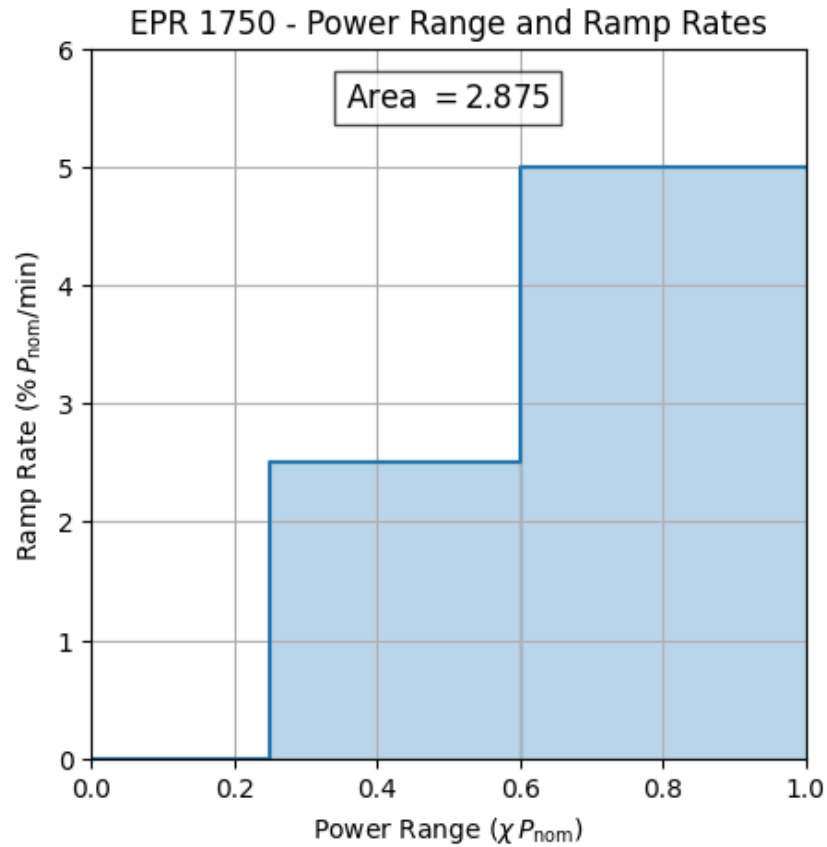


Figure 2.1: Example load following capability curve for EPR-1750

Reactor Model	Load Following Capability $\left[\frac{\%P_{nom}^2}{min} \right]$	Data Source
APR1400	0.21	[52]
EPR 1750	2.88	[82]
AP1000	4.25	[52]
BWRX-300	0.25	[52]
Rolls Royce SMR	2.00	[52]
NuScale	2.00	[83]
GE 9HA.02 2xCC	7.02	[77]
GE LM2500XPRESS+G5 2xCC	40.54	[78]
Hydroelectric	>100	[79, 80, 81]

Table 2.9: Load following capabilities and power output

3 | Analysis

4 | Results

Bibliography

- [1] James Chadwick. “The Existence of a Neutron”. In: *Proceedings of the Royal Society of London Series A* 136 (1932), pp. 692–708. DOI: 10.1098/rspa.1932.0112. URL: <https://doi.org/10.1098/rspa.1932.0112>.
- [2] Otto Hahn and Fritz Strassmann. “Nachweis der Entstehung aktiver Bariumisotope aus Uran und Thorium durch Neutronenbestrahlung; Nachweis weiterer aktiver Bruchstücke bei der Uranspaltung”. In: *Naturwissenschaften* 27.6 (Feb. 1939), pp. 89–95. ISSN: 1432-1904. DOI: 10.1007/BF01488988. URL: <https://doi.org/10.1007/BF01488988>.
- [3] Dwight D. Eisenhower. *Atoms for Peace: Address to the General Assembly of the United Nations*. Speech. Accessed: 2025-09-30. New York, NY, Dec. 1953. URL: <https://www.eisenhowerlibrary.gov/research/online-documents/atoms-peace>.
- [4] International Atomic Energy Agency (IAEA). *Power Reactor Information System (PRIS) Database*. Accessed: 2025-09-30. International Atomic Energy Agency, 2025. URL: <https://pris.iaea.org/PRIS/home.aspx>.
- [5] FundingUniverse. *Électricité de France History*. Accessed: 2025-09-30. FundingUniverse, 2025. URL: <https://www.fundinguniverse.com/company-histories/electricit%C3%A9-de-france-history/>.
- [6] Ministero dell’Interno (Italy). *Risultati del Referendum - 8 Novembre 1987*. Election Data. Accessed: 2025-09-30. Ministero dell’Interno, 1987. URL: <https://elezionistorico.interno.gov.it/index.php?tpel=F&dtel=08/11/1987&tpa=I&tpe=A&lev0=0&levsut0=0&es0=S&ms=S>.
- [7] Ministero dell’Interno (Italy). *Risultati del Referendum - 12 Giugno 2011*. Election Data. Accessed: 2025-09-30. Ministero dell’Interno, 2011. URL: <https://elezionistorico.interno.gov.it/index.php?tpel=F&dtel=12/06/2011&es0=S&tpa=Y&lev0=0&levsut0=0&ms=S&tpe=A>.
- [8] NEA. *Unlocking Reductions in the Construction Costs of Nuclear*. Accessed: 2025-09-10. Paris, 2020. URL: https://www.oecd-nea.org/jcms/pl_30653/unlocking-reductions-in-the-construction-costs-of-nuclear.

- [9] Bent Flyvbjerg. “What you Should Know about Megaprojects and Why: An Overview”. In: *Project Management Journal* 45.2 (2014), pp. 6–19. DOI: 10.1002/pmj.21409. URL: <https://doi.org/10.1002/pmj.21409>.
- [10] Naomi J. Brookes and Giorgio Locatelli. “Power plants as megaprojects: Using empirics to shape policy, planning, and construction management”. In: *Utilities Policy* 36 (2015), pp. 57–66. ISSN: 0957-1787. DOI: <https://doi.org/10.1016/j.jup.2015.09.005>. URL: <https://www.sciencedirect.com/science/article/pii/S0957178715300217>.
- [11] Jarosław Brodny and Magdalena Tutak. “Assessing the energy security of European Union countries from two perspectives – A new integrated approach based on MCDM methods”. In: *Applied Energy* 347 (2023), p. 121443. ISSN: 0306-2619. DOI: <https://doi.org/10.1016/j.apenergy.2023.121443>. URL: <https://www.sciencedirect.com/science/article/pii/S0306261923008073>.
- [12] Julianna Sterling, Missaka Warusawitharana, and Xiangyu Zhang. “European Energy Import Dependency”. In: *FEDS Notes* (Apr. 2025). DOI: 10.17016/2380-7172.3768. URL: <https://doi.org/10.17016/2380-7172.3768>.
- [13] Lasse Boehm and Alex Wilson. *EU energy security and the war in Ukraine: From sprint to marathon*. Briefing. Accessed: 2025-10-15. Feb. 2023. URL: [https://www.europarl.europa.eu/RegData/etudes/BRIE/2023/739362/EPRS_BRI\(2023\)739362_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2023/739362/EPRS_BRI(2023)739362_EN.pdf).
- [14] European Commission. *2050 long-term strategy*. https://climate.ec.europa.eu/eu-action/climate-strategies-targets/2050-long-term-strategy_en. Accessed: 2025-10-01. 2024. URL: https://climate.ec.europa.eu/eu-action/climate-strategies-targets/2050-long-term-strategy_en.
- [15] European Commission. *Commission Delegated Regulation (EU) 2021/2139 of 4 June 2021 supplementing Regulation (EU) 2020/852 of the European Parliament and of the Council by establishing the technical screening criteria for determining the conditions under which an economic activity qualifies as contributing substantially to climate change mitigation or climate change adaptation and for determining whether that economic activity causes no significant harm to any of the other environmental objectives*. Accessed: 2025-10-01. European Union, 2025. URL: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:02021R2139-20250108>.
- [16] U.S. Department of Energy (DOE). *Advanced Reactor Demonstration Program*. Accessed: 2025-10-15. U.S. Department of Energy, Office of Nuclear Energy, 2020. URL: <https://www.energy.gov/ne/advanced-reactor-demonstration-program>.

- [17] European Network of Transmission System Operators for Electricity (ENTSO-E). *Transparency Platform*. Accessed: 2025-09-30. ENTSO-E, 2025. URL: <https://transparency.entsoe.eu/>.
- [18] IFOP. *Les Français et le nucléaire : adhésion et traits d'image*. Accessed: 2025-09-30. IFOP, Sept. 2022. URL: <https://www.ifop.com/article/les-francais-et-le-nucleaire-adhesion-et-traits-dimage/>.
- [19] International Energy Agency (IEA). *Switzerland, France, Poland and Italy: Electricity Profile*. Web Page. Accessed: 2025-10-16. International Energy Agency, 2024. URL: <https://www.iea.org/countries/>.
- [20] World Nuclear Association. *Switzerland's Nuclear Industry Profile*. Accessed: 2025-09-30. World Nuclear Association, 2025. URL: <https://world-nuclear.org/information-library/country-profiles/countries-o-s/switzerland>.
- [21] European Nuclear Society (Euronuclear). *Opinion Poll: The Majority of the Swiss Support Nuclear Energy*. Accessed: 2025-09-30. European Nuclear Society, 2023. URL: <https://www.euronuclear.org/news/opinion-poll-the-majority-of-the-swiss-support-nuclear-energy/>.
- [22] NucNet. “Majority Continues to Support Nuclear Energy, Says Survey”. In: (Mar. 2023). Accessed: 2025-09-30. URL: <https://www.nucnet.org/news/majority-continues-to-support-nuclear-energy-says-survey-3-1-2023>.
- [23] World Nuclear News. “Poll Finds Majority of Swiss Support Nuclear New Build”. In: (2023). Accessed: 2025-09-30. URL: <https://www.world-nuclear-news.org/articles/poll-finds-majority-of-swiss-support-nuclear-new-build>.
- [24] SWI swissinfo.ch. “Risks Remain as Switzerland Weighs a Return to Nuclear Power”. In: (2025). Accessed: 2025-09-30. URL: <https://www.swissinfo.ch/eng/emissions-reduction/risks-remain-as-swiss-weigh-a-return-to-nuclear-power/89549444>.
- [25] NucNet. “Poland Survey”. In: (Dec. 2024). Accessed: 2025-09-30. URL: <https://www.nucnet.org/news/more-than-92-support-plans-for-first-nuclear-power-plant-12-5-2024>.
- [26] Westinghouse Electric Company. “Poland’s First Nuclear Plant Moves Forward with Westinghouse-Bechtel-PEJ Agreement”. In: (2025). Accessed: 2025-09-30. URL: <https://info.westinghousenuclear.com/news/polands-first-nuclear-plant-moves-forward-with-westinghouse-bechtel-pej-agreement>.
- [27] World Nuclear News. “Go-Ahead for Preliminary Works at Polish Plant Site”. In: (Sept. 2025). Accessed: 2025-09-30. URL: <https://www.world-nuclear-news.org/articles/go-ahead-for-preliminary-works-at-polish-plant-site>.

- [28] Electricity Maps. *2024 Carbon Intensity Data for Poland, Switzerland, France, and Northern Italy*. Accessed: 2025-09-30; Data version: January 27, 2025. Electricity Maps, Jan. 2025. URL: <https://www.electricitymaps.com>.
- [29] Il Sole 24 Ore. “Sustainable Nuclear Comes into Law: Here Is the Government Strategy in Ten Questions and Answers”. In: (2024). Accessed: 2025-09-30. URL: <https://en.ilsole24ore.com/art/sustainable-nuclear-comes-law-here-is-government-strategy-ten-questions-and-answers-AGvZZsDD>.
- [30] IPSOS. *Gli italiani e l’energia: Forum QualEnergia 2024*. Survey Report. Accessed: 2025-09-30. Legambiente, 2024. URL: <https://www.legambiente.it/wp-content/uploads/2024/11/Forum-QualEnergia-sondaggio-ipsos-2024.pdf>.
- [31] SWG S.p.A. *Nucleare italiano per i cittadini, le imprese e il territorio*. Survey Report. Accessed: 2025-09-30. iWeek, 2024. URL: <https://primapavia.it/media/2024/04/Nucleare-italiano-per-i-cittadini-le-imprese-e-il-territorio-2024-04-15.pdf>.
- [32] Bernard Roy. “Decision-aid and decision-making”. In: *European Journal of Operational Research* 45.2 (1990). OR for Engineers Expert Systems and Decision-Aid, pp. 324–331. ISSN: 0377-2217. DOI: [https://doi.org/10.1016/0377-2217\(90\)90196-I](https://doi.org/10.1016/0377-2217(90)90196-I). URL: <https://www.sciencedirect.com/science/article/pii/037722179090196I>.
- [33] Murat Köksalan, Jyrki Wallenius, and Stanley Zionts. “An Early History of Multiple Criteria Decision Making”. In: *Multiple Criteria Decision Analysis: State of the Art Surveys*. Ed. by Salvatore Greco, Matthias Ehrgott, and José Rui Figueira. New York, NY: Springer New York, 2016, pp. 3–17. ISBN: 978-1-4939-3094-4. DOI: [10.1007/978-1-4939-3094-4_1](https://doi.org/10.1007/978-1-4939-3094-4_1). URL: https://doi.org/10.1007/978-1-4939-3094-4_1.
- [34] Marco Cinelli et al. “How to support the application of multiple criteria decision analysis? Let us start with a comprehensive taxonomy”. In: *Omega* 96 (2020), p. 102261. ISSN: 0305-0483. DOI: <https://doi.org/10.1016/j.omega.2020.102261>. URL: <https://www.sciencedirect.com/science/article/pii/S0305048319310710>.
- [35] Fridolin Haag, Alice H. Aubert, and Judit Lienert. “ValueDecisions, a web app to support decisions with conflicting objectives, multiple stakeholders, and uncertainty”. In: *Environmental Modelling & Software* 150 (2022), p. 105361. ISSN: 1364-8152. DOI: <https://doi.org/10.1016/j.envsoft.2022.105361>. URL: <https://www.sciencedirect.com/science/article/pii/S1364815222000676>.
- [36] Mohammed M. Damoom et al. “Potential areas for nuclear power plants siting in Saudi Arabia: GIS-based multi-criteria decision making analysis”. In: *Progress in*

- Nuclear Energy* 110 (2019), pp. 110–120. ISSN: 0149-1970. DOI: <https://doi.org/10.1016/j.pnucene.2018.09.018>. URL: <https://www.sciencedirect.com/science/article/pii/S0149197018302464>.
- [37] Murat Başeğmez. “Strategic multi-criteria framework for nuclear plant siting: Integrating AHP, EWM, and Game Theory with GIS”. In: *Progress in Nuclear Energy* 188 (2025), p. 105897. ISSN: 0149-1970. DOI: <https://doi.org/10.1016/j.pnucene.2025.105897>. URL: <https://www.sciencedirect.com/science/article/pii/S0149197025002951>.
- [38] Ade Gafar Abdullah et al. “The social factors considered in site selection for nuclear power plants: A case study of Borneo Island, Indonesia”. In: *Progress in Nuclear Energy* 180 (2025), p. 105626. ISSN: 0149-1970. DOI: <https://doi.org/10.1016/j.pnucene.2025.105626>. URL: <https://www.sciencedirect.com/science/article/pii/S0149197025000241>.
- [39] Aravind Devanand, Markus Kraft, and Iftekhar A Karimi. “Optimal site selection for modular nuclear power plants”. In: *Computers and Chemical Engineering* 125 (2019), pp. 339–350. ISSN: 0098-1354. DOI: <https://doi.org/10.1016/j.compchemeng.2019.03.024>. URL: <https://www.sciencedirect.com/science/article/pii/S0098135419300997>.
- [40] Lauren Kiser and Luis Daniel Otero. “Multi-criteria decision model for selection of nuclear power plant type”. In: *Progress in Nuclear Energy* 159 (2023), p. 104647. ISSN: 0149-1970. DOI: <https://doi.org/10.1016/j.pnucene.2023.104647>. URL: <https://www.sciencedirect.com/science/article/pii/S0149197023000823>.
- [41] Giorgio Locatelli and Mauro Mancini. “A framework for the selection of the right nuclear power plant”. In: *International Journal of Production Research* 50.17 (2012), pp. 4753–4766. DOI: 10.1080/00207543.2012.657965. URL: <https://international.anl.gov/training/materials/BL/Literature/Giorgio%20Locatelli%20Papers/A%20framework%20for%20the%20selection%20of%20the%20right%20nuclear%20power%20plant.pdf>.
- [42] S.A. Birikorang et al. “Choosing the right reactor technology: A key step to strengthen nuclear security in West Africa’s nuclear energy plans”. In: *Annals of Nuclear Energy* 226 (2026), p. 111837. ISSN: 0306-4549. DOI: <https://doi.org/10.1016/j.anucene.2025.111837>. URL: <https://www.sciencedirect.com/science/article/pii/S0306454925006541>.
- [43] International Atomic Energy Agency (IAEA). *Application of Multi-criteria Decision Analysis Methods to Comparative Evaluation of Nuclear Energy System Options: Final Report of the INPRO Collaborative Project KIND*. Technical Report.

- Accessed: 2025-09-30. International Atomic Energy Agency, 2019. URL: https://www-pub.iaea.org/MTCD/Publications/PDF/P1853_web.pdf.
- [44] José Antonio Hernández-Torres et al. “Integrated multi-criteria decision-making approach for power generation technology selection in sustainable energy systems”. In: *Renewable Energy* 243 (2025), p. 122481. ISSN: 0960-1481. DOI: <https://doi.org/10.1016/j.renene.2025.122481>. URL: <https://www.sciencedirect.com/science/article/pii/S0960148125001430>.
 - [45] Guzay Pasaoglu, Nicolas Pardo Garcia, and Ghassan Zubi. “A multi-criteria and multi-expert decision aid approach to evaluate the future Turkish power plant portfolio”. In: *Energy Policy* 119 (2018), pp. 654–665. ISSN: 0301-4215. DOI: <https://doi.org/10.1016/j.enpol.2018.04.044>. URL: <https://www.sciencedirect.com/science/article/pii/S030142151830260X>.
 - [46] Burcin Atilgan and Adisa Azapagic. “An integrated life cycle sustainability assessment of electricity generation in Turkey”. In: *Energy Policy* 93 (2016), pp. 168–186. ISSN: 0301-4215. DOI: <https://doi.org/10.1016/j.enpol.2016.02.055>. URL: <https://www.sciencedirect.com/science/article/pii/S030142151630091X>.
 - [47] Edgar Santoyo-Castelazo and Adisa Azapagic. “Sustainability assessment of energy systems: integrating environmental, economic and social aspects”. In: *Journal of Cleaner Production* 80 (2014), pp. 119–138. ISSN: 0959-6526. DOI: <https://doi.org/10.1016/j.jclepro.2014.05.061>. URL: <https://www.sciencedirect.com/science/article/pii/S0959652614005381>.
 - [48] Muhammed Zulfakar Zolkaffly and Ki-In Han. “Reactor technology assessment and selection utilizing systems engineering approach”. In: *AIP Conference Proceedings* 1584.1 (Feb. 2014), pp. 22–31. ISSN: 0094-243X. DOI: 10.1063/1.4866099. eprint: https://pubs.aip.org/aip/acp/article-pdf/1584/1/22/11739325/22_1_online.pdf. URL: <https://doi.org/10.1063/1.4866099>.
 - [49] Fatih Topaloğlu. “Analytic network process (ANP) based decision support tool for nuclear power plant location and reactor type selection”. In: *Nuclear Engineering and Technology* 57.3 (2025), p. 103228. ISSN: 1738-5733. DOI: <https://doi.org/10.1016/j.net.2024.09.031>. URL: <https://www.sciencedirect.com/science/article/pii/S1738573324004777>.
 - [50] Rahul Goushis. “Evaluating the feasibility of modular nuclear reactors in the European Union: A GIS-based multicriteria decision analysis approach to meet Paris Agreement”. In: *Applied Energy* 400 (2025), p. 126565. ISSN: 0306-2619. DOI: <https://doi.org/10.1016/j.apenergy.2025.126565>. URL: <https://www.sciencedirect.com/science/article/pii/S0306261925012954>.

- [51] *Nuclear Power Reactors in the World*. entity 2. Vienna: INTERNATIONAL ATOMIC ENERGY AGENCY, 2024. ISBN: 978-92-0-122224-4. URL: [/publications/15748/nuclear-power-reactors-in-the-world](#).
- [52] International Atomic Energy Agency. *Advanced Reactors Information System (ARIS)*. Accessed: 2025-06-10. 2025. URL: <https://aris.iaea.org/>.
- [53] KEPCO Engineering and Inc. Construction Company. *Reactor Descriptions*. Accessed: 2025-06-10. 2025. URL: <https://www.kepco-enc.com/menu.es?mid=a20202030100>.
- [54] A. N. Author. *Example Reference Title*. Accessed: 2025-09-10. City, 2025. URL: <https://www.example.com>.
- [55] World Nuclear News. *Early works agreement for BWRX-300 SMRs in Finland and Sweden*. Accessed: 2025-06-10. 2025. URL: <https://www.world-nuclear-news.org/articles/early-works-agreement-for-bwr-300-smrs-in-finland-and-sweden>.
- [56] World Nuclear News. *Hungary looking at deployment of US small modular reactors*. Accessed: 2025-06-10. 2025. URL: <https://www.world-nuclear-news.org/articles/hungary-looking-at-deployment-of-us-small-modular-reactors>.
- [57] TerraPower. *Sodium Reactor Technology*. Accessed: 2025-06-10. 2025. URL: <https://www.terrapower.com/sodium/>.
- [58] World Nuclear News. *South Korean SMR design approved by regulator*. Accessed: 2025-06-10. Sept. 26, 2024. URL: <https://www.world-nuclear-news.org/articles/south-korean-smr-design-approved-by-regulator>.
- [59] U.S. Nuclear Regulatory Commission (NRC). *NRC Approves Standard Design for NuScale US460 Small Modular Reactor*. News Release. Accessed: 2025-10-16. May 2025. URL: <https://www.nrc.gov/cdn/doc-collection-news/2025/25-033.pdf>.
- [60] Congressional Research Service (CRS). *Considerations for Reprocessing of Spent Nuclear Fuel*. Report R48364. Accessed: 2025-10-16. Congressional Research Service, 2025. URL: <https://www.congress.gov/crs-product/R48364>.
- [61] M. Arslan et al. “MOX recycling in GEN 3 + EPR Reactor homogeneous and stable full MOX core”. In: American Nuclear Society, 555 North Kensington Avenue, La Grange Park, IL 60526 (United States). American Nuclear Society - ANS; La Grange Park (United States), July 2013. URL: <https://www.osti.gov/biblio/22257897>.
- [62] Robert J. Fetterman. “AP1000 core design with MOX loading”. In: *Annals of Nuclear Energy* 36.3 (2009). PHYSOR 2008, pp. 324–330. ISSN: 0306-4549. DOI:

- <https://doi.org/10.1016/j.anucene.2008.11.022>. URL: <https://www.sciencedirect.com/science/article/pii/S0306454908002983>.
- [63] Canadian Nuclear Safety Commission. *BWRX-300 Design: Non-Proprietary Information on Fuel Design Qualification and BWR Fuel Licensing*. Technical Report. Accessed: 2025-09-30. Canadian Nuclear Safety Commission, 2024. URL: <https://api.cnscccsn.gc.ca/dms/digital-medias/CMD24-H3-Ref-BWRX-300-DNNP-GNF2-Fuel-Design-Qualification-and-BWR-Fuel-Licensing-Non-Proprietary-Information.PDF>.
 - [64] Rolls-Royce SMR. *Generic Design Assessment (GDA) Public Consultation: Have Your Say*. Accessed: 2025-09-30. Rolls-Royce SMR, 2025. URL: <https://gda.rolls-royce-smr.com/have-your-say>.
 - [65] U.S. Nuclear Regulatory Commission. *Standard Review Plan for Environmental Reviews for Nuclear Power Plants*. Technical Report. Accessed: 2025-09-30. U.S. Nuclear Regulatory Commission, 2017. URL: <https://www.nrc.gov/docs/ML1700/ML17007A001.pdf>.
 - [66] NuScale Power. *NuScale Announces MOX Capability*. Accessed: 2025-09-30. NuScale Power, Jan. 2016. URL: <https://www.nuscalepower.com/press-releases/2016/nuscale-announces-mox-capability>.
 - [67] Geoffrey Rothwell. “Projected electricity costs in international nuclear power markets”. In: *Energy Policy* 164 (2022), p. 112905. ISSN: 0301-4215. DOI: <https://doi.org/10.1016/j.enpol.2022.112905>. URL: <https://www.sciencedirect.com/science/article/pii/S0301421522001306>.
 - [68] NEA. *Unlocking Reductions in the Construction Costs of Nuclear*. Paris: OECD Publishing, 2020.
 - [69] U.S. Energy Information Administration (EIA). *Capital Cost and Performance Characteristics for Utility Scale Power Generating Technologies*. Technical Report. Accessed: 2025-10-16. U.S. Energy Information Administration, Jan. 2024. URL: https://www.eia.gov/analysis/studies/powerplants/capitalcost/pdf/capital_cost_AE02025.pdf.
 - [70] Björn Steigerwald et al. “Uncertainties in estimating production costs of future nuclear technologies: A model-based analysis of small modular reactors”. In: *Energy* 281 (2023), p. 128204. ISSN: 0360-5442. DOI: <https://doi.org/10.1016/j.energy.2023.128204>. URL: <https://www.sciencedirect.com/science/article/pii/S0360544223015980>.
 - [71] Reuters Events. “GE Hitachi chases gas plant displacement with new 300 MW reactor”. In: *Reuters Events / Nuclear* (Apr. 2018). Accessed: 2025-10-01. URL:

- <https://www.reutersevents.com/nuclear/ge-hitachi-chases-gas-plant-displacement-new-300-mw-reactor>.
- [72] Power Technology. “Rolls Royce plans first UK modular nuclear reactor for 2029”. In: *Power Technology* (Apr. 2022). Accessed: 2025-10-01. URL: <https://www.power-technology.com/news/uk-first-smr-rolls-royce/>.
 - [73] NuScale Power. *NuScale Power Announces an Additional 25 Percent Increase in NuScale Power Module Output*. Accessed: 2025-10-01. Nov. 2020. URL: <https://www.nuscalepower.com/press-releases/2020/nuscale-power-announces-an-additional-25-percent-in-nuscale-power-module-output>.
 - [74] GE Vernova. *Nuclear Energy Solutions*. Accessed: 2025-09-30. GE Vernova, 2025. URL: <https://www.gevernova.com/nuclear>.
 - [75] Rolls-Royce SMR. *Our Approach to Small Modular Reactors*. Accessed: 2025-09-30. Rolls-Royce SMR, 2025. URL: <https://www.rolls-royce-smr.com/our-approach>.
 - [76] NuScale Power. *NuScale Power Module: Product Overview*. Accessed: 2025-09-30. NuScale Power, 2025. URL: <https://www.nuscalepower.com/products/nuscale-power-module>.
 - [77] General Electric (GE). *GE 9HA Gas Turbine: Technical Overview and Performance Specifications*. Technical Brochure. Accessed: 2025-10-16. 2015. URL: <https://f.machineryhost.com/1c549272d486597e78d102330c36981e/USPE%20GE%209HA%20Gas%20Turbine.pdf>.
 - [78] GE Vernova. *LM2500XPRESS Gas Turbine: Product Factsheet*. Product Factsheet. Accessed: 2025-10-16. 2025. URL: https://www.gevernova.com/content/dam/gepower-new/global/en_US/downloads/gas-new-site/products/gas-turbines/gev-aero-fact-sheets/GEA35746-GEV-LM2500XPRESS-Product-Factsheet.pdf.
 - [79] U.S. Congress, Office of Technology Assessment. *Electric Power Wheeling and Dealing: Technological Considerations for Increasing Competition*. Report OTA-E-409. Washington, DC: U.S. Government Printing Office, May 1989. URL: <https://ota.fas.org/reports/8913.pdf>.
 - [80] P F Shiers, F S Fisher, and J A Kosberg. “The Mt. Hope Waterpower Project: A pumped storage facility that enhances the environment while supplying on-demand electrical needs”. In: American Society of Civil Engineers, New York, NY (United States), Dec. 1995. URL: <https://www.osti.gov/biblio/219975>.
 - [81] MWH. *Technical Analysis of Pumped Storage and Integration with Wind Power in the Pacific Northwest*. Technical Report. Accessed: 2025-10-16. U.S. Army Corps of Engineers Northwest Division Hydroelectric Design Center, Aug. 2009. URL:

- <https://www.hydro.org/wp-content/uploads/2017/08/PS-Wind-Integration-Final-Report-without-Exhibits-MWH-3.pdf>.
- [82] OECD Nuclear Energy Agency (NEA). *Technical and Economic Aspects of Load Following with Nuclear Power Plants*. Technical Report. Accessed: 2025-09-30. OECD Nuclear Energy Agency, June 2011. URL: https://www.oecd-nea.org/upload/docs/application/pdf/2021-12/technical_and_economic_aspects_of_load_following_with_nuclear_power_plants.pdf.
 - [83] ENCO. *Analysis of Small Modular Reactor (SMR) Concepts*. Technical Report. Accessed: 2025-09-30. Federal Ministry of Labour and Economy (BMLUK), Austria, 2022. URL: https://www.bmluk.gv.at/dam/jcr:26805e08-47f2-402e-8121-921e1f121144/Analysis_of_SMR_Concepts.pdf.
 - [84] U.S. Nuclear Regulatory Commission. *Design Description*. Technical Report. Accessed: 2025-09-30. U.S. Nuclear Regulatory Commission, 2005. URL: <https://www.nrc.gov/docs/ML0522/ML052280170.pdf>.
 - [85] Rolls-Royce SMR. *Rolls-Royce SMR Select UK-Based Fuel Solution*. Accessed: 2025-09-30. Rolls-Royce SMR, Oct. 2023. URL: <https://www.rolls-royce-smr.com/press/rolls-royce-smr-select-uk-based-fuel-solution>.
 - [86] Framatome. *GNF2 Fuel Assembly: Product Sheet*. Accessed: 2025-09-30. Framatome, May 2018. URL: https://www.framatome.com/solutions-portfolio/docs/default-source/default-document-library/product-sheets/a0529-p-us-g-en-328-05-18-http.pdf?Status=Master&sfvrsn=60b7ce1d_2.
 - [87] U.S. Nuclear Regulatory Commission. *Safety Evaluation by the Office of Nuclear Reactor Regulation: NuScale Power Module*. ML23161A411. Accessed: 2025-09-30. U.S. Nuclear Regulatory Commission, June 2011. URL: <https://www.nrc.gov/docs/ML1117/ML11171A411.pdf>.
 - [88] AREVA NP, EDF Energy. *Sub-Chapter 15.1: Level 1 Probabilistic Safety Assessment*. Technical Report. Accessed: 2025-09-30. EDF Energy, Nov. 2012. URL: https://www.edfenergy.com/sites/default/files/sub-chapter_15.1_-_level_1_psa.pdf.
 - [89] Korea Hydro and Nuclear Power Co., Ltd. (KHNP). *Major Characteristics of the EU-APR Reactor*. Presentation. Accessed: 2025-09-30. Apr. 2018. URL: <https://sep.com.pl/photo/files/07b%20-%20KHNP%20-%20Major%20Characteristics%20of%20EU-APR%20Reactor.pdf>.
 - [90] H. Paillere. “Nuclear energy for a net zero world”. In: (2021), [12 p.]
 - [91] International Atomic Energy Agency (IAEA). *Climate Change and Nuclear Power 2022*. Report. Accessed: 2025-09-30. International Atomic Energy Agency, 2022.

- URL: <https://www.iaea.org/topics/nuclear-power-and-climate-change/climate-change-and-nuclear-power-2022>.
- [92] Bojie Liu et al. “Critical review of nuclear power plant carbon emissions”. In: *Frontiers in Energy Research* Volume 11 - 2023 (2023). ISSN: 2296-598X. DOI: 10.3389/fenrg.2023.1147016. URL: <https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2023.1147016>.
- [93] Travis S. Carless, W. Michael Griffin, and Paul S. Fischbeck. “The environmental competitiveness of small modular reactors: A life cycle study”. In: *Energy* 114 (2016), pp. 84–99. ISSN: 0360-5442. DOI: <https://doi.org/10.1016/j.energy.2016.07.111>. URL: <https://www.sciencedirect.com/science/article/pii/S0360544216310350>.
- [94] Our World in Data. *Death Rates from Energy Production per TWh*. Accessed: 2025-09-30; Data sources: Markandya & Wilkinson (2007); Sovacool et al. (2016); UN-SCEAR (2008 & 2018). Our World in Data, 2025. URL: <https://ourworldindata.org/grapher/death-rates-from-energy-production-per-twh>.
- [95] Anil Markandya and Paul Wilkinson. “Electricity Generation and Health”. In: *The Lancet* 370.9591 (Sept. 2007), pp. 979–990. ISSN: 0140-6736. DOI: 10.1016/S0140-6736(07)61253-7. URL: [https://doi.org/10.1016/S0140-6736\(07\)61253-7](https://doi.org/10.1016/S0140-6736(07)61253-7).
- [96] Benjamin K. Sovacool et al. “Balancing safety with sustainability: assessing the risk of accidents for modern low-carbon energy systems”. In: *Journal of Cleaner Production* 112 (2016), pp. 3952–3965. ISSN: 0959-6526. DOI: <https://doi.org/10.1016/j.jclepro.2015.07.059>. URL: <https://www.sciencedirect.com/science/article/pii/S0959652615009877>.
- [97] National Council on Radiation Protection and Measurements (NCRP). *Ionizing Radiation Exposure of the Population of the United States*. Report 160. Paywalled; Data accessed via U.S. EPA summary. National Council on Radiation Protection and Measurements, 2009. URL: <https://ncrponline.org/publications/reports/ncrp-report-160-2/>.
- [98] U.S. Environmental Protection Agency (EPA). *Radiation Sources and Doses*. Accessed: 2025-10-15; Data sourced from NCRP Report No. 160 (2009). U.S. Environmental Protection Agency, 2025. URL: <https://www.epa.gov/radiation/radiation-sources-and-doses>.
- [99] J. P. McBride et al. “Radiological Impact of Airborne Effluents of Coal and Nuclear Plants”. In: *Science* 202.4372 (Dec. 1978), pp. 1045–1050. ISSN: 0036-8075. DOI: 10.1126/science.202.4372.1045. URL: <https://doi.org/10.1126/science.202.4372.1045>.

- [100] Ralph L. Keeney. “Value-focused thinking: Identifying decision opportunities and creating alternatives”. In: *European Journal of Operational Research* 92.3 (1996), pp. 537–549. ISSN: 0377-2217. DOI: [https://doi.org/10.1016/0377-2217\(96\)00004-5](https://doi.org/10.1016/0377-2217(96)00004-5). URL: <https://www.sciencedirect.com/science/article/pii/S0377221796000045>.
- [101] Paul Goodwin and George Wright. *Decision Analysis for Management Judgment*. Wiley, 2004. ISBN: 978-0-470-86108-0. URL: <https://books.google.ch/books?id=yxPgmAEACAAJ>.
- [102] A Morton and B Fasolo. “Behavioural decision theory for multi-criteria decision analysis: a guided tour”. In: *Journal of the Operational Research Society* 60.2 (2009), pp. 268–275. DOI: 10.1057/palgrave.jors.2602550. URL: <https://doi.org/10.1057/palgrave.jors.2602550>.
- [103] Peter Reichert et al. “The conceptual foundation of environmental decision support”. In: *Journal of Environmental Management* 154 (2015), pp. 316–332. ISSN: 0301-4797. DOI: <https://doi.org/10.1016/j.jenvman.2015.01.053>. URL: <https://www.sciencedirect.com/science/article/pii/S0301479715001140>.
- [104] Mika Marttunen, Valerie Belton, and Judit Lienert. “Are objectives hierarchy related biases observed in practice? A meta-analysis of environmental and energy applications of Multi-Criteria Decision Analysis”. In: *European Journal of Operational Research* 265.1 (2018), pp. 178–194. ISSN: 0377-2217. DOI: <https://doi.org/10.1016/j.ejor.2017.02.038>. URL: <https://www.sciencedirect.com/science/article/pii/S0377221717301777>.
- [105] Alice H. Aubert, Fabien Esculier, and Judit Lienert. “Recommendations for online elicitation of swing weights from citizens in environmental decision-making”. In: *Operations Research Perspectives* 7 (2020), p. 100156. ISSN: 2214-7160. DOI: <https://doi.org/10.1016/j.orp.2020.100156>. URL: <https://www.sciencedirect.com/science/article/pii/S2214716020300464>.
- [106] International Energy Agency (IEA). *Energy Statistics Data Browser: Electricity Imports vs. Exports, France, 1990–2024*. Data Set. Accessed: 2025-10-16; Licence: CC BY 4.0. International Energy Agency, 2024. URL: <https://www.iea.org/data-and-statistics/data-tools/energy-statistics-data-browser?country=FRA&fuel=Imports%2Fexports&indicator=ElecImportsExports>.

A | Appendix A

If you need to include an appendix to support the research in your thesis, you can place it at the end of the manuscript. An appendix contains supplementary material (figures, tables, data, codes, mathematical proofs, surveys, . . .) which supplement the main results contained in the previous chapters.

B | Appendix B

It may be necessary to include another appendix to better organize the presentation of supplementary material.

List of Figures

2.1	Example load following capability curve for EPR-1750	21
-----	--	----

List of Tables

2.1	Selection of reactor models for analysis	12
2.2	Selection of advanced reactor models for analysis	13
2.3	Final selection of reactor models for analysis	14
2.4	MOX fuel compatibility scores and maximum MOX fraction	16
2.5	Options overview	17
2.6	Capital costs for selected reactor models	18
2.7	Technological readiness and planned commissioning time	18
2.8	Economic costs for selected reactor models	19
2.9	Load following capabilities and power output	21

List of Symbols

Variable	Description	SI unit
$\boldsymbol{u}$	solid displacement	m
$\boldsymbol{u}_f$	fluid displacement	m

Acknowledgements

Here you might want to acknowledge someone.

This work was supported by the ENEN2plus mobility grant.



**Co-funded by
the European Union**

